Arthur Pesah

Postdoctoral researcher at FU Berlin

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Academic Positions

March 2026–Now **Freie Universität (FU) Berlin**, *Postdoctoral researcher.*
Working on quantum error error correction in the group of Jens Eisert.

Education

2021–2025 **University College London**, *PhD*, Quantum Computing.
Supervisor: Dan Browne
PhD student in Dan Browne's group.
Thesis: *Quantum error correction in three dimensions and beyond*

2020–2021 **University College London**, *Master of Research*, Quantum Technologies.
Master of Research (MRes) funded by the Center for Doctoral Training (CDT) in Delivering Quantum Technologies.
Thesis: *Tailoring 3D topological codes and decoders to biased noise*

2017–2019 **KTH Royal Institute of Technology**, *Master of Science*, Theoretical Physics.
Double-degree with my engineering's school. I took courses in particle physics, statistical physics, general relativity, quantum computing, dynamical systems, and algebraic topology.
Thesis: *Learning quantum state properties with quantum and classical neural networks*

2015–2017 **ENSTA Paris**, *Engineering's degree*, Applied Mathematics.
One of the top French "Grandes Écoles". I took courses in applied mathematics (statistics, optimization, numerical methods, functional analysis) and physics (quantum physics, relativity, statistical physics, fluid/solid mechanics, plasma).

2013–2015 **Lycée Henri IV**, *Preparatory Class*, Mathematics and Physics.
A French specific two years' undergraduate program leading to a nation-wide competitive examination into a "Grande École", one of the top French engineers school system. Henri IV: Top 3 preparatory classes.

Research Experience

November 2022-Now **Research consultant**, *Xanadu Quantum Technologies.*
I work eight hours a week as a contractor for Xanadu, helping design an architecture for their photonic quantum computers.

April-July 2025 **Student Researcher**, *Google*, Santa Barbara, California, USA.
Advisor: Craig Gidney
Internship at Google Quantum AI, in the quantum error correction theory team.

Jan.-April 2024 **Research intern**, *Xanadu Quantum Technologies*, Toronto, Canada.
Advisors: Michael Vasmer and Ilan Tzitrin
Full-time internship with Xanadu, in the quantum error correction team. Resulted in a paper: *Fault-tolerant transformations of spacetime codes.*

June-Sept. 2020 **Summer Student**, *Los Alamos National Laboratory*, Los Alamos, New Mexico, USA.
Advisors: Marco Cerezo and Patrick Coles
Student and researcher at the Quantum Computing Summer School organized by LANL. Resulted in a paper: *Absence of Barren Plateau in Quantum Convolutional Neural Networks*

March-May. 2020 **Research Assistant**, *DTU Technical University of Denmark*, Copenhagen, Denmark.
Advisor: Lars Kai Hansen
I worked on two projects in the computer science department of DTU: federated learning for EEG data and thermal state preparation with noisy quantum circuits. Resulted in a paper: *Noise-Assisted Variational Quantum Thermalization*

June-Dec. 2019 **Machine Learning Scientist**, *1QBit*, Vancouver and Waterloo, Canada.
Advisor: Pooya Ronagh
Member of the Hardware Innovation Lab, a research team that explores how future quantum devices can be leveraged to solve practical industrial problems. I designed reinforcement learning algorithms to control nonstoquastic quantum annealers. Summer in Vancouver and Fall in Waterloo.

Oct. 2018 - May 2019 **Visiting Graduate Student**, *University of Toronto*, Toronto, Canada.
Advisors: Peter Wittek (University of Toronto) and Nathan Killoran (Xanadu).
Internship in the context of my master's thesis, I worked on learning quantum state properties with quantum neural networks (for both continuous and discrete states)

July-Sept. 2018 **Research Engineer**, *University of Liège*, Liège, Belgium.
Advisor: Gilles Louppe.
I worked for the summer on deep learning and meta-learning methods for likelihood-free inference. Resulted in a paper: *Recurrent Machines for Likelihood-free Inference*

Sept. 2017 - June 2018 **Research Assistant**, *KTH Royal Institute of Technology*, Stockholm, Sweden.
Advisor: Hossein Azizpour
Research assistantship in *Robotics, Perception and Learning (RPL)* lab. I worked on benchmarking adversarial domain adaptation methods.

June - Aug. 2017 **Summer Student**, *CERN*, Geneva, Switzerland.
Advisors: Benoît Salvant and Nicolo Biancacci
Member of the CERN Summer Student Program, I contributed to the development of a new model of instability inside particle accelerators.

Sept. 2015 - June 2017 **Research Assistant**, *Université Paris-Saclay*, Orsay, France.
Advisor: Isabelle Guyon
I worked on domain adaptation with High Energy physics datasets. I developed a software that enables to test and visualize domain adaptation algorithms on toy datasets.

Teaching

June - Aug. 2022 **UCLQ Undergraduate Summer Program**, *University College London*.
I supervised a summer project on implementing a Union-Find decoder for the 3D surface code, with two undergraduate students.

Jan. - May 2022 **Quantum Computation and Communication (PHAS0070)**, *University College London*.
Teaching assistant for the postgraduate course on quantum computing at UCL.

Jan. 2020 **PSI Winter School**, *Perimeter Institute*.
I supervised a project on the loss landscape of variational circuits with three master students, at a Winter School organized by the Perimeter Institute.

Oct. 2018 - May 2019 **Quantum Machine Learning MOOC**, *University of Toronto*.
Teaching Assistant for the QML MOOC (available on EdX) made by Peter Wittek. I wrote some of the lecture notebooks and helped answering questions on the forum.

March - May 2018 **Deep Learning in Data Science (DD2424)**, *KTH Royal Institute of Technology*.
Teaching Assistant for the deep learning master's course at KTH. I helped students with their projects and homework and was in charge of setting up Google Cloud Platform for the course.

July - Aug. 2015 **AutoML Hackathons**, *ChaLearn*.
Teaching assistant for the non-profit organization ChaLearn during two hackathons on automatic machine learning (AutoML). One was in Saint-Petersburg, at the *Machine Learning and Intelligence School* and the other in Stanford at the *INNS Conference on Big Data*. I wrote the starting kit for the challenge and helped the participants during the hackathons

Papers

2025 **A. Pesah**, D. Austin, I. Tzitrin, M. Vasmer, *Fault-tolerant transformations of spacetime codes*, arXiv:2509.09603

2025 Xanadu collaboration, *Scaling and networking a modular photonic quantum computer*, Nature

2024 T. Scruby, **A. Pesah**, M. Webster, *Quantum rainbow codes*, arXiv:2408.13130

2023 K. Tiurev, **A. Pesah**, P.-J. H. S. Derks, J. Roffe, J. Eisert, M. S. Kesselring, J.-M. Reiner, *The domain wall color code*, Physical Review Letters

2022 **A. Pesah***, E. Huang*, C. Chubb, M. Vasmer, A. Dua, *Tailoring Three-Dimensional Surface Codes for Biased Noise*, PRX Quantum (*same authorship position)

2021 J. Foldager, **A. Pesah** and L. K. Hansen, *Noise-Assisted Variational Quantum Thermalization*, Nature Scientific Reports

2020 **A. Pesah**, M. Cerezo, S. Wang, T. Volkoff, A.T. Sornborger, P.J. Coles, *Absence of Barren Plateaus in Quantum Convolutional Neural Networks*, Physical Review X

2020 W. Guan, G. Perdue, **A. Pesah**, M. Schuld, K. Terashi, S. Vallecorsa, J. Vlimant, *Quantum Machine Learning in High Energy Physics*, Machine Learning: Science and Technology

2018 **A. Pesah**, A. Wehenkel, G. Louppe, *Recurrent Machines for Likelihood-free Inference*, NeurIPS 2018 Workshop on Meta-Learning

Scientific blog posts

2023 A. Pesah, *Computing with quantum codes using transversal gates*, personal website

2023 A. Pesah, *An interactive introduction to the surface code*, personal website

2023 A. Pesah, *Quantum error correction with the stabilizer formalism*, personal website

2022 A. Pesah, *Introduction to classical error correction*, personal website

2022 A. Pesah, *A bird's-eye view of quantum error correction and fault tolerance*, personal website

2018 A. Pesah and A. Wehenkel, *Improve your scientific models with meta-learning and likelihood-free inference*, Towards Data Science

2018 A. Pesah, *Recent Advances for a Better Understanding of Deep Learning*, Towards Data Science

2018 A. Pesah, *A Little Review of Domain Adaptation in 2017*, personal website

Projects

2021- A. Pesah and E. Huang, *PanQEC*, open-source quantum error correction library for the simulation and 3D visualization of topological codes (web visualizer available at `gui.quantumcodes.io`)

Talks

2026 **Contributed talk**, *QCTIP 2026*, Oxford, UK.
Fault-tolerant transformation of spacetime codes

2025 **Contributed talk**, *QEC 2025*, Yale, USA.
Fault-tolerant transformation of spacetime codes (available on Yale Media Library)

2025 **Invited talk**, *YITP Logical Gates for Encoded Qubits Workshop*, Kyoto, Japan.
Fault-tolerant transformations of spacetime codes (available on YouTube)

2025 **Seminar talk**, *FU Berlin*, Berlin, Germany.
Fault-tolerant transformation of spacetime codes

2024 **Invited talk**, *TOPO 2023*, Sydney, Australia.
Towards a grand unified theory of biased noise

2024 **Conference talk**, *FTQC 2024*, Benasque, Spain.
Quantum Rainbow Codes

2024 **Seminar talk**, *Perimeter Institute*, Waterloo, Canada.
Quantum Rainbow Codes (available on pirsa.org)

2023 **Seminar talk**, *Riverlane*.
Quantum error correction with PanQEC

2022 **Seminar talk**, *FU Berlin*, Berlin, Germany.
Tailoring three-dimensional surface codes for biased noise (available on YouTube)

2022 **Conference talk**, *APS March Meeting 2022*.
Tailoring three-dimensional surface codes for biased noise

2021 **Conference talk**, *APS March Meeting 2021*.
Absence of Barren Plateaus in Quantum Convolutional Neural Networks (slides available on my website)

2020 **Seminar talk**, *IRISA (CNRS) and DTU Compute*, France and Denmark.
Quantum Machine Learning Beyond the Hype (slides available on my website)

2019 **Seminar talk**, *IQC Student Seminar, University of Waterloo*, Waterloo, Canada.
A Gradient-based Method for Controlling Adiabatic Trajectories

2018 **Contributed talk**, *NeurIPS 2018 Workshop on Meta-Learning*.
Recurrent Machines for Likelihood-free Inference

2018 **Summer School Lecture**, *2nd International Summer School on Deep Learning*.
Representation Learning, Domain Adaptation and Generative Models with Deep Learning (lecture on adversarial domain adaptation)

Grants and Awards

2019 **Mitacs Globalink Research Award**, *6,000 CAD*.
Canadian grant received for my master's thesis internship at the University of Toronto

2017 **ENSTA Best Research Project Award**, *2nd position*, 1,000€.
Award received for my research internship at CERN during the Summer 2017. Competition organized by ENSTA ParisTech to gratify the best research summer internships. Students are judged by a dozen researchers of the school on a 50-pages report, the comments of the supervisor and a 3 minutes presentation. 16 students are selected in the first phase (among the whole cohort consisting in 150 students) and 3 prizes are awarded at the end.

Community service

Reviewer for journals and conferences.

- *ACM Symposium on Theory of Computing (STOC 2026)*
- *Quantum Information Theory in Practice (QCTIP 2024)*
- *Quantum Information Processing conference (QIP 2023, 2024)*
- *Theory of Quantum Computation, Communication, and Cryptography conference (TQC 2022)*
- *Quantum Journal*
- *PRX*, *PRX Quantum*, *PRA*, American Physical Society
- *Journal of Physics Communication*, IOP Publishing
- *Machine Learning and the Physical Sciences 2019 workshop*, NeurIPS 2019

2021 **Program Committee Member**, *6th International Conference for Young Quantum Information Scientists*, Michigan State University, April 2021.
Helping in the organization of the conference (reviewing, chairing sessions, finding invited speakers, etc.)

Other activities

2024-2025 **Science mentoring**, *CITY SPARC Project*, London.
Mentoring of three high-school students for a quantum computing project done in the context of a program to expose young students to scientific research

2016 **Science outreach**, *Bouge la Science*, Supelec, Gif-sur-Yvettes.
Popularized scientific experiments to junior high school students during a scientific festival in the engineering school Supelec.

2016 **Science outreach**, *Palais de la Decouverte*, Paris.
Volunteered to present some physics experiments to a general public at a science museum in Paris

2015 **French selection for the International Physicists Tournament (IPT)**, *3rd place*.
Member of the ENSTA team, I participated to the national selection of the IPT. We worked on open physics problems during 4 months and presented models and experiments in front of a jury.

2015 **Founder and President**, *DaTA - ENSTA Computer Science Club*, Palaiseau.
Founder of my engineering's school computer science club. We organized formations (web development, GitHub, machine learning, etc.), helped students and professors with their computer problems, and made projects together

2015 **Co-Founder**, *GraviTAtion - ENSTA Physics Club*, Palaiseau.
Co-Founder of my engineering's school physics club. We organized the Interational Physicists Tournament preparation in our school, as well as some seminars and science outreach events

Skills

Langages	Python, C/C++, JavaScript
Quantum Libraries	PanQEC, Stim, Qiskit, Cirq, PennyLane
ML Libraries	Tensorflow, PyTorch, Keras, Scikit-Learn
Web Framework	Three.js